

Chapter 27: Recent advances in rare-earth free permanent magnet materials for future device applications

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Abstract

People have been studying permanent magnets (PMs) for a long time, even before the term “materials science” existed. Due to the unique properties and various applications, the development of PMs has completely changed modern technology. The need for rare-earth-free (REF) PMs has drawn a lot of attention due to the rising cost and depleting reserves of rare-earth (RE)-base PMs. The materials with $L1_0$ -ordered tetragonal structures have excellent magnetic properties. The high magnetic anisotropy of the $L1_0$ structure originates from its chemically long-range ordered arrangement of alternating monoatomic layers. $L1_0$ -ordered magnetic materials have therefore attracted a lot of attention in the production of strong non-RE PMs. $L1_0$ -ordered magnetic materials, including Fe-based alloys (FePt, FeNi), CoPt, and Mn-based compounds (MnAl and MnGa), exhibit outstanding magnetic performance, making them suitable for permanent magnet applications. This chapter discusses the $L1_0$ -ordered rare-earth free (REF) permanent magnets (PMs) materials and their future applications.

Keywords: Permanent magnets, $L1_0$ -type magnetic materials, rare-earth free materials, magnetic anisotropy

1. Introduction

It is widely accepted that permanent magnets are crucial parts of modern energy and technological systems. Modern technology like Audio Equipment, computer hard disk drives (HDDs), Motors and Generators, Medical Devices all make extensive use of permanent magnets (PMs)[1][2]. In systems such as hard disk drives (HDD), the magnetization of permanent magnets can offer a stable mass storing of data, which is another important use in information technologies. Permanent magnets have most recently been used in newly developed nanoscale devices like magnetic random-access memory (MRAM)[3], [4], [5], [6], [7], [8]. The most notable feature of permanent magnets is their capacity to maintain a high magnetic flux in the application environment without an imposed magnetic field, which makes them crucial for the applications mentioned above. The maximal energy product $(BH)_{\max}$ (in J/m^3 or MGOe), this provides an approximation of the potential magnetic flux density stored in the magnet, is the quality factor of permanent magnet's. High thermal stability and high value of $(BH)_{\max}$ permanent magnets are much sought after for sophisticated applications. The production of high-performance magnets has advanced significantly over the past century with the discovery of rare-earth (RE) PMs like $SmCo_5$, Sm_2Co_{17} , and $Nd_2Fe_{14}B$ [1], [9], [10]. For example, $Nd_2Fe_{14}B$ magnets have energy products up to 400 kJ m^{-3} (50 MGOe), which is an order of magnitude greater than those of conventional ferrite or Alnico magnets[11], [12]. Despite their benefits, there are significant supply issues with RE-based magnets. The manufacturing of permanent magnets with superior performance, which are utilized in almost all contemporary technologies, requires RE elements, specially Nd, Dy, and Tb[13], [14], [15], [16], [17], [18]. On the other hand, only a few nations produce RE materials, and China is the main producer. The detrimental effects of mining and processing RE elements on the environment, in addition to their geological shortage, are other important concerns [19], [20], [21], [22]. In context, researchers were investigated several non-rare earth materials to find exceptional magnetic characteristics. In the early 1900s researchers discovered $L1_0$ -ordered non RE materials, shows extraordinary magnetic properties which are useful for PM applications. Fe-Pt, CoPt, and Mn-based MnAl, MnGa, and other $L1_0$ -ordered

2. L1₀-type magnetic materials

3. L1₀-MnAl

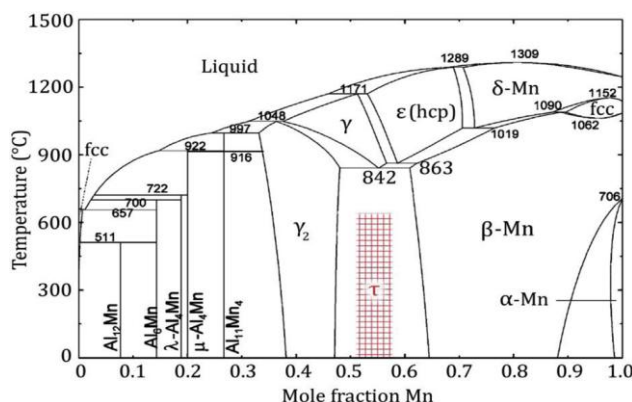


Fig.1. Phase diagram of MnAl system. Reproduced from [29] with permission from Springer Nature. Copyright© 2016 Springer Nature.

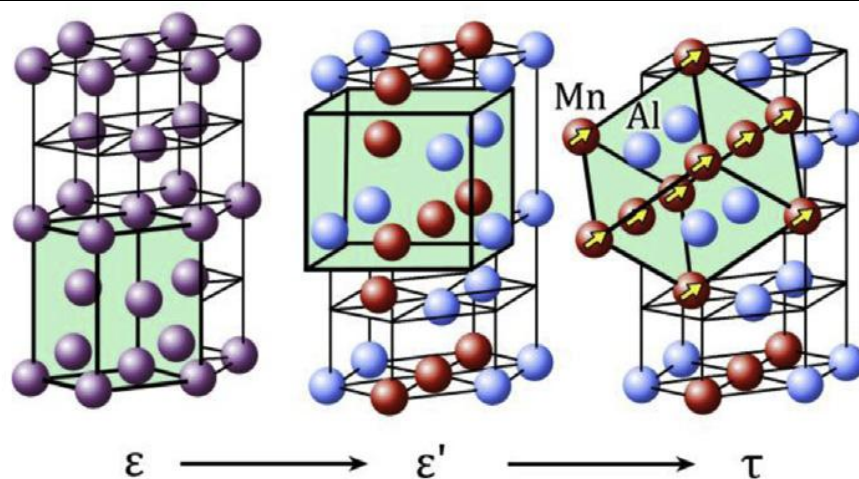


Fig. 2. Diagrammatic depiction of the lattice relationship between the ϵ , ϵ' , and $L1_0 \tau$ phases with closely spaced planes oriented parallel. Reproduced from [29] with permission from Elsevier. Copyright © 1979 Elsevier.

Within the $P4/mmm$ space group, τ -phase MnAl crystallizes like the $L1_0$ -structure and displays hard ferromagnetic characteristics[30]. The τ -phase MnAl possesses a large magnetocrystalline anisotropy energy of 1.525 MJ/m^3 , highlighting its strong potential for use in high-performance permanent magnets[31]. The MnAl shows τ -phase when the Mn composition range between 51–58.5 at%[32]. Along with ferromagnetic τ - phase MnAl shows some additional anti-ferromagnetic γ_2 -MnAl phase with a Ideal composition of Mn_5Al_8 and β -Mn phase at $\text{Mn} > 58.5 \text{ at\%}$ [33], [34]. Fig.1. shows the phase diagram of MnAl system. Preparation of τ - Mn-Al alloys is a difficult task because the presence of anti-ferromagnetic γ_2 -MnAl and β -Mn phase, which affects the ferromagnetism of τ - Mn-Al. Therefore, most suggested way to synthesis τ - Mn-Al is derived from the anti-ferromagnetic ϵ -MnAl phase. When MnAl anneal at above 870°C followed by rapid cooling, which form ϵ -MnAl phase. Then annealing with temperature range between 350 to 650°C causes the ϵ to τ -MnAl phase transformation[35], [36], [37]. Van den Broek et al. investigated that the ϵ -phase experiences a diffusional reordering into an intermediate ϵ' -phase upon chilling at a medium cooling rate of $10\text{--}20 \text{ K/min}$. Then additional shear mechanism that converts the ϵ -phase with an $L1_0$ structure into the τ -phase. Fig.2. Diagrammatic depiction of the lattice relationship between the ϵ , ϵ' , and $L1_0 \tau$ phases. Fang et al. theoretically analyzed the magnetic properties of MnAl systems, they have demonstrated relatively high magneto-crystalline anisotropy of 17 Merg/cm^3 (1.7 MJ/m^3), $T_C = 650 \text{ K}$, and maximum energy product $(BH)_{\text{max}} = 12.4 \text{ MGOe}$ (98.7 kJ/m^3)[38]. Thus, it has good potential for permanent magnet applications. P. Saraswathi et al. prepared powder $\text{Mn}_{54}\text{Al}_{46}$ through heat-free high-energy ball milling, they obtained magnetic properties of $H_c = 3.0 \text{ kOe}$, $M_r = 0.05 \text{ emu/g}$, $M_s = 0.19 \text{ emu/g}$, and $(BH)_{\text{max}} = 432 \text{ J/m}^3$ (0.054 MGOe) [39]. Mn_xAl ($0.7 \leq x \leq 3$) films deposited on GaAs(001) substrate with CoGa underlayer at 200° deposition temperature then annealed at 200°C for 20 min, obtained coercivity of 0.22 to 5.38 T and $(BH)_{\text{max}} = 52.18 \text{ kJ/m}^3$ (6.56 MGOe) Hongli Sun et al.[40]. According to reports, C-doped MnAl alloy has improved machinability, mechanical qualities, and structural stability. The $\text{Mn}_{54}\text{Al}_{46}$ alloy prepared using arc-melting, melt-spinning, and post-heat-treatment processes without carbon exhibited promising magnetic properties at room temperature, including a saturation magnetization of 650.5 kA/m , a coercivity of 0.5 T , and a $(BH)_{\text{max}}$ value of 24.7 kJ/m^3 (3.10 MGOe). The Curie temperature of $\text{Mn}_{54-x}\text{Al}_{46}\text{C}_x$ alloys decreases with carbon incorporation, while the saturation magnetization is retained at a comparable level.

4. L1₀-type MnGa

Hames et al. initially reported the ferromagnetic phase of MnGa in 1948[42]. The MnGa shows L1₀-phase when the Mn composition range between 65–74.5 at%[43]. When single crystal Mn_{1-x}Ga_x (x=62±2 %) thin films grown on GaAs form thermodynamically stable δ -phase (L10) reported by Kannan[44]. Further Tanaka et al. use Molecular Beam Epitaxy (MBE) on GaAs substrates to create the stable ferromagnetic phase Mn_xGa_{1-x} (x=0.55–0.60).[45]. When L1₀-Mn_{1.5}Ga films grown on GaAs(100) substrate, the film produced coercivity of 42.8 kOe and (BH)_{max} of 2.60 MGOe (20.7 kJ/m³) at room temperature[46]. Recent study shows that by adding third element like Al, Pt and C able to tune the magnetic characteristics of L1₀-MnGa[47], [48]. Substitution of Al and C can tune the lattice parameter c[47]. Pt doping on L1₀-MnGa can increase the out-of-plane coercivities. The coercivity of the Mn₆₀Ga₄₀ alloy was 441.08 kA/m. Coercivity peaked at 1172.77 kA/m with a 15% Pt content[48]. Overall research indicates that the L10-MnGa alloy's atomic makeup and manufacturing method have a major influence on its magnetic characteristics.

5. L1₀-type FePt and CoPt

The study of the novel magnetic materials is of great interest among the researchers owing to their unique magnetic properties towards the fulfilment of current demands for technological advancements. Out of different classes of magnetic materials, 3d transition metal based ferromagnetic alloys are gaining a kind of special interest, as their extensive applications spanning primarily from energy generation to harvesting inevitably fulfil our everyday essential needs [49], [50]. The partially filled 3d-electrons in ferromagnetic Fe, Ni and Co, i.e. the presence of unpaired electrons in their valence shell contribute to the overall spin magnetic moments, making them responsible for their observable magnetic behavior. Because of their superior structural and magnetic qualities, both theoretical and experimental investigations on alloying transition metals with platinum (Pt) have garnered a lot of interest in the last two decades[51], [52], [53], [54], [55], and ordered binary FePt and CoPt has been identified as promising hard ferromagnetic alloys over other contemporary materials. Moreover, FePt and CoPt alloys are among those potential candidates towards the development of rare-earth free permanent magnets and continuously expanding their territory for numerous spintronic specific device applications[52], [53], [54], [55], [56].

Platinum-based alloys, however, outperform traditional rare-earth magnets in a number of different ways. These are-

1. FePt and CoPt are well known for their extraordinary magnetic characteristics. Their special L1₀ crystal structure, which has the maximum magnetic anisotropy, keeps them from rapidly demagnetizing at the nanoscale level, where other magnets would fail miserably.
2. Conventional magnets entirely lose their magnetism at sizes smaller than 10 nm because of temperature fluctuations, while FePt and CoPt can retain permanent magnetism even at these sizes.
3. High temperatures cause conventional NdFeB magnets to lose their magnetism, while FePt and CoPt's exceptional heat tolerance allows them to retain their thermal stability and magnetic characteristics.
4. Rare-earth metals like neodymium and dysprosium are extremely sensitive, prone to severe supply chain chokepoints and price volatility, whereas platinum is pricey. Developing alloys based on platinum avoids these geopolitical problems.

Structural Property

Fe and Co being 3d transition metal ferromagnets, while platinum, one of the most precious silvery-white 5d transition metals, widely recognized for its remarkable resistance to corrosion, high melting point, chemically inert along with other excellent physical and chemical properties. FePt and CoPt are very closely related binary transition metal-Pt alloys. They have gathered particular attention from the research community due to their intrinsically very high magnetocrystalline anisotropy energy (MAE), arising from the strong spin-orbit interaction of Pt with Fe or Co atoms [50], [57]. By definition, MAE is the dependence of a material's magnetic property with respect to its crystallographic directions and accordingly a magnetic material can have easy (requires lower energy to reach up to saturation magnetization) and hard axis (higher amount of energy is required for saturation) directions within the crystal. At ordinary temperatures, FePt (or CoPt) alloy thin films don't possess enough thermal energy for the formation of ordered structure, and hence they typically form a disordered face-centered cubic structure (fcc). This disordered structure transforms into a highly ordered face-centered tetragonal (fct or L_{10} phase) structure followed by high temperature processing such as heating the substrate during deposition and/or post annealing of the room temperature deposited sample which assist to get past the energy barrier that prevents atomic diffusion [49], [50], [51]. It enables the disordered arrangement of Fe (or Co) and Pt atoms to rearrange themselves in layered stacks directed along c-axis of the lattice followed by establishment and stabilization of the L_{10} ordered FePt (or CoPt) phase. During chemical ordering in the L_{10} phase, the interplay between strong spin-orbit coupling agent Pt and hybridization of Pt 5d bands with Fe or Co 3d bands results in a huge value of MAE constant [54], [55]. This phase transformation breaks the cubic symmetry, the a-axis of FePt (or CoPt) lattice expands and shrinkage of the c-axis facilitates tetragonal lattice formation, which can be experimentally verified by X-ray diffraction (XRD) analysis of ordered and disordered alloy phases [57].

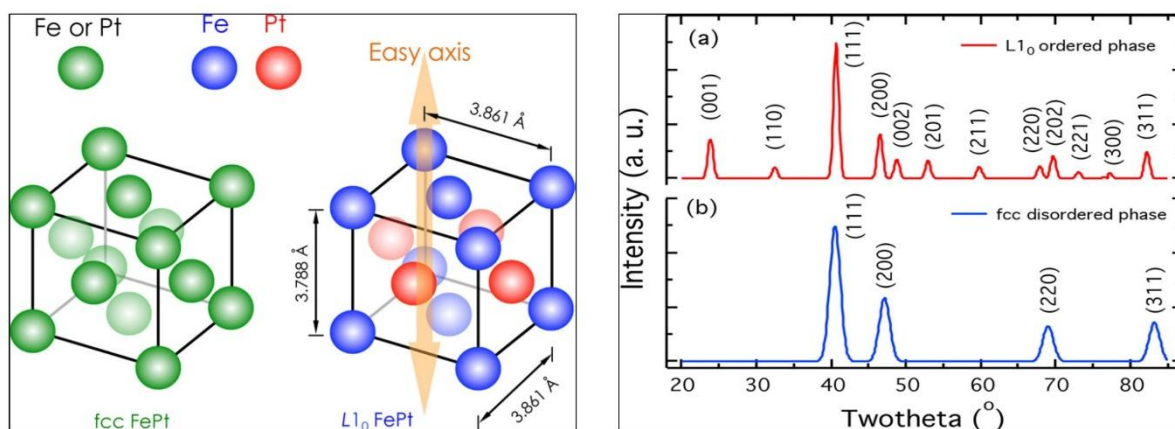


Fig.3. Crystal structure of FePt alloy in the ordered (L_{10} fct) and disordered (fcc) state (left image) and corresponding x-ray diffraction patterns (right image). Reproduced from [57] with permission from Elsevier. Copyright © 2016 Elsevier.

Table-1: structural and magnetic characteristics of FePt and CoPt alloys[50].

Material	at. % of Pt	Space group	Structure	a = b (nm)	c (nm)	Ku (10^7 erg/cc)	Ms (emu/c)	Tc (K)
FePt	50	Fm3m	A1 fcc	0.381	0.381	-	-	-
FePt	50	P4/mmm	L1 ₀ fct	0.3853	0.3713	6.6	1140	750
CoPt	50	Fm3m	A1 fcc	0.3768	0.3768	-	-	-
CoPt	50	P4/mmm	L1 ₀ fct	0.3803	0.3701	4.9	800	840

Magnetic Property

The ordered-disordered phases of the crystalline structure of these alloys directly reflects in their magnetic properties. The atomic composition of the individual element, film thickness, and substrate type are some of the considerable factors that affect the disorder–order transition temperature. Although, FePt (or CoPt) possesses similar crystal structure, exhibit large spin-orbit coupling due to Pt, they differ greatly in terms of magnetic parameters. This difference comes from the magnetic moment contribution from Fe, Co and Pt atoms, the contribution to total spin moments from Pt atoms is much less compared to Fe and Co atoms in all the binary transition metal-Pt alloys systems[53]. A summary table comprising the comparison of structural and magnetic characteristics of FePt and CoPt alloys has been listed in table 1. The fcc phase of FePt (or CoPt) is characterized by soft magnetic nature and usually less preferred from application viewpoint, while fct phase emerges with high value of coercivity ($H_c \sim 20\text{kOe}$), saturation magnetization ($M_s \sim 1100\text{ emu/cc}$), uniaxial MAE, ($K_u \sim 10^7\text{ erg/cc}$) and high Curie temperature ($T_c \sim 750\text{K}$) which are essential for practical applications in permanent magnets and data storage industry. In experiments, K_u seems to be highly dependent on the degree of chemical order, which reveals the excellence of the L1₀ phase. The high K_u values of these alloy stabilizes the thermal fluctuations by overcoming superparamagnetic effect and enable to obtain particle size reduced to $\sim 4\text{nm}$, an ultimate necessity for achieving high areal density of stored bits ($\sim 10\text{ Tb/in}^2$) in future recording media such as heat assisted magnetic recording (HAMR)[58]. Also, high T_c value of FePt and CoPt alloys makes them suitable for HAMR media design [57].

Applications

A recent study on L1₀ FePt thin films has demonstrated its potential application as a cover layer for MFM (Magnetic Force Microscope) tips. They have reported an enormous coercivity of 13.2 kOe in 20 nm thin Fe₅₁Pt₄₉ thin films and then practically used it to record distinct, high resolution MFM images[59]. Another study on L1₀FePt granular thin film with hexagonal boron nitride (hBN) as a grain boundary material investigates the microstructural as well as magnetic properties for HAMR (Heat Assisted Magnetic Recording) applications. The authors have reported successful fabrication of 16 nm thin FePt-hBN granular media allowing well-isolated and high aspect ratio magnetic grains with superior microstructure and magnetic properties.

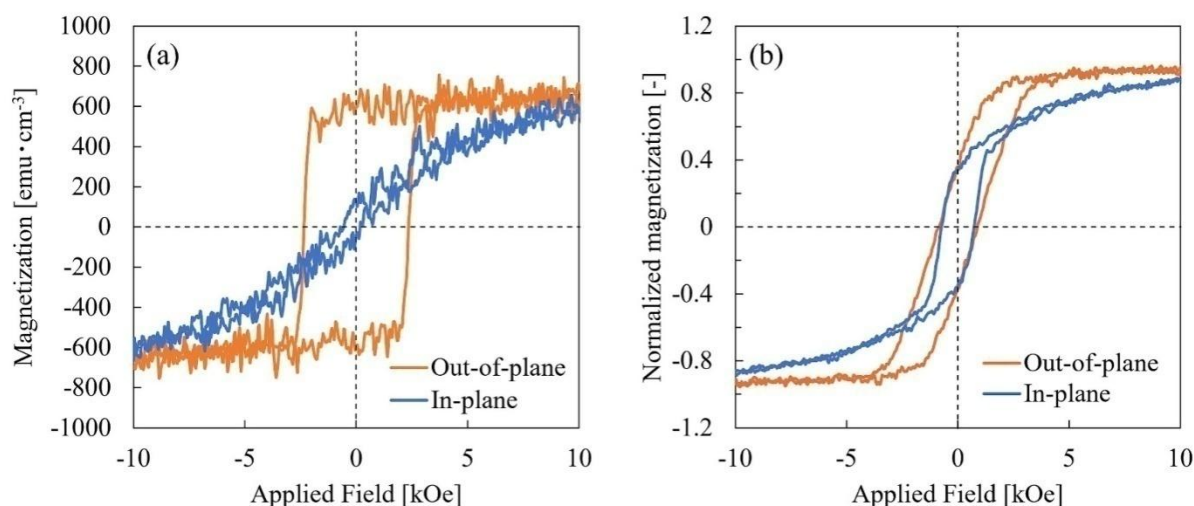


Fig.4. Magnetization curves of electrodeposited 5 nm ultrathin CoPt films along both in-plane and out-of-plane directions. Reproduced from [61] with permission from Elsevier. Copyright © 2025 Elsevier.

On the other hand, Suzuki et al. has observed perpendicular magnetic anisotropy (PMA) in extremely thin (~ 3 nm) CoPt films using electrodeposition technique for 3D domain-wall motion memory (3D-DWMM) recording layer applications[60]. Similar studies on 5 nm ultrathin CoPt film manifests high PMA ($K_u \sim 4.33\text{Merg/cc}$) and coercivity as high as 2.4 kOe [61]. The produced CoPt films hold great potential for the creation of next-generation magnetic memories with ultrahigh recording densities and magnetic tunnel junctions (MTJs) with improved tunnel magnetoresistance effects. More recently, spin-orbit torque (SOT) driven magnetization switching measurements conducted using CoPt alloy films without the support of an externally applied magnetic field emerges as a hot topic of research in scientific community. The single layer CoPt film has showed its ability to behave as current induced perpendicular magnetization switching where magnetization flipping has been observed with an in-plane electrical current which can generate sufficient torque. For low-power non-volatile memory and logic devices, it is a basic prerequisite. This has been observed in a compositionally graded CoPt single layers, where internal inversion symmetry breaking from a composition gradient enables field-free SOT switching of perpendicular magnetization at room temperature without the need for an external magnetic field [56].

6. Conclusion

The production of RE-free permanent magnets offers an appropriate path to resource-resilient and sustainable magnetic materials. In this chapter, we have discussed an overview of the fundamental principles and important properties of L1₀-type magnetic materials. However, none of the L1₀-type magnetic materials have yet attained the theoretical maximum energy product, which is a crucial standard for practical practicality. Achieving superior performance at a reduced cost, particularly by lowering the cost per unit of (BH)_{max} to match or outperform existing REE magnets, is the key priority. Given the performance of the materials and the market demand, rare-earth permanent magnets will continue to play a significant role in permanent magnet applications for the foreseeable future, despite advancements in the quest for new rare-earth-free magnets.

References:

- [1]. Coey, J. M. D. (2011). Hard magnetic materials: A perspective. *IEEE Transactions on magnetics*, 47(12), 4671-4681. doi: 10.1109/TMAG.2011.2166975.
- [2]. Coey, J. M. D. (2012). Permanent magnets: Plugging the gap. *Scripta Materialia*, 67(6), 524-529. doi: 10.1016/j.scriptamat.2012.04.036.
- [3]. Eshraghian, K. (2010). Evolution of nonvolatile resistive switching memory technologies: the related influence on heterogeneous nanoarchitectures. *Transactions on Electrical and Electronic Materials*, 11(6), 243-248. doi: 10.4313/TEEM.2010.11.6.243.
- [4]. Yuasa, S., Nagahama, T., Fukushima, A., Suzuki, Y., & Ando, K. (2004). Giant room-temperature magnetoresistance in single-crystal Fe/MgO/Fe magnetic tunnel junctions. *Nature materials*, 3(12), 868-871. <https://www.nature.com/articles/nmat1257>
- [5]. Wang, L., Yang, C. H., & Wen, J. (2015). Physical principles and current status of emerging non-volatile solid state memories. *Electronic materials letters*, 11(4), 505-543. doi: 10.1007/s13391-015-4431-4.
- [6]. Park, S., Bae, J. K., Park, S. H., Kwon, H. R., & Jang, H. W. (2025). Rare-Earth-Free Iron-Based Permanent Magnets: Progress, Challenges, and Perspectives. *MetalMat*, 2(4), e70022. doi: 10.1002/metm.70022.
- [7]. Bhatti, S., Sbiaa, R., Hirohata, A., Ohno, H., Fukami, S., & Piramanayagam, S. N. (2017). Spintronics based random access memory: a review. *Materials today*, 20(9), 530-548. doi: 10.1016/j.mattod.2017.07.007.
- [8]. Chappert, C., Fert, A., & Van Dau, F. N. (2007). The emergence of spin electronics in data storage. *Nature materials*, 6(11), 813-823.
- [9]. Sugimoto, S. (2011). Current status and recent topics of rare-earth permanent magnets. *Journal of Physics D: Applied Physics*, 44(6), 064001. doi: 10.1088/0022-3727/44/6/064001.
- [10]. Lewis, L. H., & Jiménez-Villacorta, F. (2013). Perspectives on permanent magnetic materials for energy conversion and power generation. *Metallurgical and Materials Transactions A*, 44(Suppl 1), 2-20. doi: 10.1007/s11661-012-1278-2.
- [11]. Gutfleisch, O., Willard, M. A., Brück, E., Chen, C. H., Sankar, S. G., & Liu, J. P. (2011). Magnetic materials and devices for the 21st century: stronger, lighter, and more energy efficient. *Advanced materials*, 23(7), 821-842. <https://advanced.onlinelibrary.wiley.com/doi/full/10.1002/adma.201002180>
- [12]. Parmar, H., Xiao, T., Chaudhary, V., Zhong, Y., & Ramanujan, R. V. (2017). High energy product chemically synthesized exchange coupled Nd₂Fe₁₄B/ α -Fe magnetic powders. *Nanoscale*, 9(37), 13956-13966. doi: 10.1039/C7NR02348K.

- [13]. Srinithi, A. K., Tang, X., Sepehri-Amin, H., Ohkubo, T., & Hono, K. (2024). (Nd, Sm, La)-Fe-B-based hot-deformed magnets with excellent comprehensive properties for variable flux permanent magnet motors. *Acta Materialia*, 281, 120387. doi: 10.1016/j.actamat. 2024.120387.
- [14]. Xu, Y., & Hadjipanayis, G. C. (2024). Investigation of rare earth lean Sm-(Fe, Co)-Mo alloys for permanent magnets. *Journal of Magnetism and Magnetic Materials*, 589, 171484. doi: 10.1016/j.jmmm.2023.171484.
- [15]. Gjoka, M., Sarafidis, C., & Giaremis, S. (2024). Towards production of cost-effective modification of SmCo5-type alloys suitable for permanent magnets. *Materials*, 17(4), 808. doi: 10.3390/ma17040808.
- [16]. Huang, C. C., Mo, C. C., & Ou, S. F. (2024). Influence of Nd-base alloy addition on improving the characteristics of recycled sintered NdFeB permanent magnets. *Journal of Magnetism and Magnetic Materials*, 600, 172166. doi: 10.1016/j.jmmm.2024.172166.
- [17]. Huang, C. C., & Mo, C. C. (2024). Vacuum processing temperature effect on highly coercive recycled NdFeB permanent magnets with Pr-based alloy addition. *Materials Chemistry and Physics*, 325, 129648. doi: 10.1016/j.matchemphys.2024.129648.
- [18]. Zhang, X. F., Liu, L. B., Li, Y. Q., Zhang, D. T., Liu, W. Q., & Yue, M. (2024). Preparation and magnetic hardening of low Ti content (Sm, Zr)(Fe, Co, Ti) 12 magnets by rapid solidification non-equilibrium method. *Chinese Physics B*, 33(9), 097503. doi: 10.1088/1674-1056/ad58c4.
- [19]. Bai, J., Xu, X., Duan, Y., Zhang, G., Wang, Z., Wang, L., & Zheng, C. (2022). Evaluation of resource and environmental carrying capacity in rare earth mining areas in China. *Scientific Reports*, 12(1), 6105. <https://www.nature.com/articles/s41598-022-10105-2>
- [20]. Liu, H., Guo, H., Pourret, O., Wang, Z., Liu, M., Zhang, W., ... & Laine, P. (2022). Geochemical signatures of rare earth elements and yttrium exploited by acid solution mining around an ion-adsorption type deposit: Role of source control and potential for recovery. *Science of the Total Environment*, 804, 150241. doi: 10.1016/j.scitotenv. 2021.150241.
- [21]. Su, J., Guo, X., Gao, Y., Wu, S., Xu, R., & Sun, X. (2021). Recovery of thorium and rare earths from leachate of ion-absorbed rare earth radioactive residues with N1923 and Cyanex® 572. *Journal of Rare Earths*, 39(10), 1273-1281. doi: 10.1016/j.jre.2020.09.005.
- [22]. Wang, Z., Luo, Y., Zheng, C., An, C., & Mi, Z. (2020). Spatial distribution, source identification, and risk assessment of heavy metals in the soils from a mining region: A case study of Bayan Obo in northwestern China. *Human and Ecological Risk Assessment: An International Journal*, 27(5), 1276-1295. doi: 10.1080/10807039.2020.1821350.
- [23]. Ristau, R. A., Barmak, K., Lewis, L. H., Coffey, K. R., & Howard, J. K. (1999). On the relationship of high coercivity and L1 ordered phase in CoPt and FePt thin films. *Journal of applied physics*, 86(8), 4527-4533. doi: 10.1063/1.371397.

-
- [24]. Park, J. H., Hong, Y. K., Bae, S., Lee, J. J., Jalli, J., Abo, G. S., ... & Lee, J. G. (2010). Saturation magnetization and crystalline anisotropy calculations for MnAl permanent magnet. *Journal of Applied Physics*, 107(9). doi: 10.1063/1.3337640.
- [25]. Wang, K., Lu, E., Knepper, J. W., Yang, F., & Smith, A. R. (2011). Structural controlled magnetic anisotropy in Heusler L1– MnGa epitaxial thin films. *Applied Physics Letters*, 98(16). doi: 10.1063/1.3582244.
- [26]. Poirier, E., Pinkerton, F. E., Kubic, R., Mishra, R. K., Bordeaux, N., Mubarak, A., ... & Barmak, K. (2015). Intrinsic magnetic properties of L1 FeNi obtained from meteorite NWA 6259. *Journal of Applied Physics*, 117(17). doi: 10.1063/1.4916190.
- [27]. Sakuma, N., Ohshima, T., Shoji, T., Suzuki, Y., Sato, R., Wachi, A., ... & Teranishi, T. (2011). Exchange coupling interaction in L 10-FePd/ α -Fe nanocomposite magnets with large maximum energy products. *ACS nano*, 5(4), 2806-2814. <https://pubs.acs.org/doi/abs/10.1021/nn103286r>
- [28]. Odkhuu, D., & Hong, S. C. (2019). First-principles prediction of possible rare-earth free permanent magnet of tetragonal Fe Co with enhanced magnetic anisotropy and energy product through interstitial nitrogen. *Physical Review Applied*, 11(5), 054085. <https://journals.aps.org/prapplied/abstract/10.1103/PhysRevApplied.11.054085>
- [29]. Cui, J., Kramer, M., Zhou, L., Liu, F., Gabay, A., Hadjipanayis, G., ... & Sellmyer, D. (2018). Current progress and future challenges in rare-earth-free permanent magnets. *Acta Materialia*, 158, 118-137.
- [30]. Koch, A. J. J., Hokkeling, P., vd Steeg, M. G., & De Vos, K. J. (1960). New material for permanent magnets on a base of Mn and Al. *Journal of Applied Physics*, 31(5), S75-S77. <https://pubs.aip.org/aip/jap/article-abstract/31/5/S75/147477/New-Material-for-Permanent-Magnets-on-a-Base-of-Mn>
- [31]. Pareti, L., Bolzoni, F., Leccabue, F., & Ermakov, A. E. (1986). Magnetic anisotropy of MnAl and MnAlC permanent magnet materials. *Journal of applied physics*, 59(11), 3824-3828. doi: 10.1063/1.336723.
- [32]. Chaturvedi, A., Yaqub, R., & Baker, I. (2014). A comparison of τ -MnAl particulates produced via different routes. *Journal of Physics: Condensed Matter*, 26(6), 064201. <https://iopscience.iop.org/article/10.1088/0953-8984/26/6/064201/meta>
- [33]. Ellner, M. (1990). The structure of the high-temperature phase MnAl (h) and the displacive transformation from MnAl (h) into Mn₅Al₈. *Metallurgical Transactions A*, 21(6), 1669-1672. <https://link.springer.com/article/10.1007/BF02672582>
- [34]. Preston, G. D. (1928). CXXV. The crystal structure of β -manganese. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 5(33), 1207-1225. <https://www.tandfonline.com/doi/abs/10.1080/14786440608564570>
-

- [35]. Crisan, A. D., Vasiliu, F., Nicula, R., Barthä, C., Mercioniu, I., & Crisan, O. (2018). Thermodynamic, structural and magnetic studies of phase transformations in MnAl nanocomposite alloys. *Materials Characterization*, 140, 1-8. doi: 10.1016/j.matchar.2018. 03.034.
- [36]. Lu, W., Niu, J., Wang, T., Xia, K., Xiang, Z., Song, Y., ... & Yan, Y. (2016). Phase transformation kinetics and microstructural evolution of MnAl permanent magnet alloys. *Journal of Alloys and Compounds*, 685, 992-996.
- [37]. Dehghan, H., Rezayat, M. and Ebrahimi, S. A. S., *Modeling of strain induced transformation during hot deformation of an Mn–Al–C alloy*, Materials Science and Engineering: A, **vol. 776**, p. 139006(2020). doi: 10.1016/j.msea.2020.139006.
- [38]. Fang, H., Kontos, S., Ångström, J., Cedervall, J., Svedlindh, P., Gunnarsson, K., & Sahlberg, M. (2016). Directly obtained τ -phase MnAl, a high performance magnetic material for permanent magnets. *Journal of Solid State Chemistry*, 237, 300-306. doi: 10.1016/j.jssc.2016.02.031.
- [39]. Saraswathi, P., Madeswaran, S., Subhavanekadesh, S., Vizhi, R. E., & Babu, D. R. (2024). Enhancement of Coercivity in α -Mn (Al) Magnetic Nanopowders Processed by High-Energy Ball Milling. *Journal of Superconductivity and Novel Magnetism*, 37(8), 1723-1731. <https://link.springer.com/article/10.1007/s10948-024-06816-4>
- [40]. Sun, H., Zhao, X., Han, R., Zhang, C., Liu, L., Qin, H., ... & Zhao, J. (2024). Compositionally modulated perpendicular magnetic anisotropy in tetragonal Mn_xAl films. *Applied Physics Letters*, 124(10). doi: 10.1063/5.0183072.
- [41]. Wei, J. Z., Song, Z. G., Yang, Y. B., Liu, S. Q., Du, H. L., Han, J. Z., ... & Yang, J. B. (2014). τ -MnAl with high coercivity and saturation magnetization. *Aip Advances*, 4(12). doi: 10.1063/1.4903773.
- [42]. Hames, F. A., & Eppelsheimer, D. S. (1948). Some new ferromagnetic manganese alloys. *Nature*, 162(4129), 968-968.
- [43]. Minakuchi, K., Umetsu, R. Y., Ishida, K., & Kainuma, R. (2012). Phase equilibria in the Mn-rich portion of Mn–Ga binary system. *Journal of Alloys and compounds*, 537, 332-337. doi: 10.1016/j.jallcom.2012.04.065.
- [44]. Krishnan, K. M. (1992). Ferromagnetic δ -Mn $1-x$ Ga x thin films with perpendicular anisotropy. *Applied physics letters*, 61(19), 2365-2367. doi: 10.1063/1.108245.
- [45]. Tanaka, M., Harbison, J. P., Sands, T. D., Philips, B. A., De Boeck, J., Cheeks, T. L., ... & Keramidas, V. G. (1993). Epitaxial Ferromagnetic MnGa and (MnNi) Ga Thin Films with Perpendicular Magnetization on GaAs. *MRS Online Proceedings Library (OPL)*, 313, 507. doi: 10.1557/PROC-313-507.
- [46]. Zhu, L., Nie, S., Meng, K., Pan, D., Zhao, J., & Zheng, H. (2012). Multifunctional L10-Mn 1.5 Ga films with ultrahigh coercivity, giant perpendicular magnetocrystalline anisotropy and large magnetic energy product. *Advanced Materials*, 24(33), 4547-4551. doi: 10.1002/adma.201200805.

- [47]. Jiao, Z., Wang, J., Fu, Z., Zhang, R., & Jiang, C. (2021). Magnetocrystalline anisotropy regulations in bulk L10-MnGa alloys by tailoring the tetragonal lattice parameter c: selectively alloying Al and C atoms. *Journal of Alloys and Compounds*, 881, 160646. doi: 10.1016/j.jallcom.2021.160646.
- [48]. Zhang, Y., Zhang, W., Ning, M., Chen, L., & Li, H. (2021). Tuning the magnetism of L10-MnGa films by Pt doping. *Applied Surface Science*, 542, 148585. doi: 10.1016/j.apsusc.2020.148585.
- [49]. Weller, D., Mosendz, O., Parker, G., Pisana, S., & Santos, T. S. (2013). L10 Fe P t X–Y media for heat-assisted magnetic recording. *physica status solidi (a)*, 210(7), 1245-1260. <https://onlinelibrary.wiley.com/doi/full/10.1002/pssa.201329106>
- [50]. Andreazza, P., Pierron-Bohnes, V., Tournus, F., Andreazza-Vignolle, C., & Dupuis, V. (2015). Structure and order in CoPt-type nanoalloys: From thin films to supported clusters. *Surf. Sci. Rep*, 70(2), 188. doi: 10.1016/j.surfrep.2015.02.002.
- [51]. Varvaro, G., Laureti, S., & Fiorani, D. (2014). L10 FePt-based thin films for future perpendicular magnetic recording media. *Journal of magnetism and magnetic materials*, 368, 415-420. doi: 10.1016/j.jmmm.2014.04.058.
- [52]. Vashisht, G., Goyal, R., Bala, M., Ojha, S., & Annapoorni, S. (2018). Studies of exchange coupling in FeCo/L1 0-FePt bilayer thin films. *IEEE Transactions on Magnetics*, 55(3), 1-5. doi: 10.1109/TMAG.2018.2876167.
- [53]. Alsaad, A., Ahmad, A. A., & Obeidat, T. S. (2020). Structural, electronic and magnetic properties of the ordered binary FePt, MnPt, and CrPt3 alloys. *Heliyon*, 6(3). [https://www.cell.com/heliyon/fulltext/S2405-8440\(20\)30390-X](https://www.cell.com/heliyon/fulltext/S2405-8440(20)30390-X)
- [54]. Aledealat, K., Aladerah, B., Obeidat, A., & Gharaibeh, M. (2021). First-principles study of electronic structure and magnetic properties of L10-ordered FeNi, FePd, and FePt alloys. *Heliyon*, 7(12). doi: 10.1016/j.heliyon.2021.e08639.
- [55]. Lethole, N. L., Mukumba, P., & Makaka, G. (2023). First-principles calculations to investigate magnetic, electronic, mechanical and dynamical properties of the bimetallic M-Pt (M: Mn Co and Ni) alloys. *Journal of Magnetism and Magnetic Materials*, 565, 170298. doi: 10.1016/j.jmmm.2022.170298.
- [56]. Liu, L., Zhou, C., Zhao, T., Yao, B., Zhou, J., Shu, X., ... & Chen, J. (2022). Current-induced self-switching of perpendicular magnetization in CoPt single layer. *Nature Communications*, 13(1), 3539.
- [57]. Hsiao, S. N. (2016). FePt thin films: Fundamentals and applications.
- [58]. Gayen, A., Saravanan, P., & Perumal, A. (2013). Magnetic Properties of Ultra-Thin FePt Films Grown on Oxidized Si Substrates. In *Advanced Nanomaterials and Nanotechnology: Proceedings of the 2nd International Conference on Advanced Nanomaterials and Nanotechnology, Dec 8–10, 2011, Guwahati, India* (pp. 457-464). Berlin, Heidelberg: Springer Berlin Heidelberg. https://link.springer.com/chapter/10.1007/978-3-642-34216-5_45

-
- [59]. Zhou, J., Han, Z., Wang, X., Zhang, L., Ma, Z., Ma, L., & Zheng, F. (2024). Microstructure and magnetic properties of FePt thin films for the coating of magnetic force microscopy tips. *Journal of Materials Science: Materials in Electronics*, 35(18), 1193.
<https://link.springer.com/article/10.1007/s10854-024-12987-6>
- [60]. Suzuki, R., Saito, M., Oshima, D., Kato, T., Sakata, K., Amemiya, K., & Homma, T. (2025). Electrodeposition and microstructural characterization of CoPt ultrathin films with perpendicular magnetic anisotropy. *Electrochimica Acta*, 523, 145972. doi: 10.1016/j.electacta.2025.145972.
- [61]. Araki, D., Sonobe, Y., Takahashi, Y. K., & Homma, T. (2025). Ultrathin CoPt alloy films with fcc (111) orientation and perpendicular magnetic anisotropy fabricated by electrodeposition. *Electrochemistry Communications*, 176, 107938. doi: 10.1016/j.elecom.2025.107938.
- [62]. Xu, C., Varaprasad, B. C. S., Laughlin, D. E., & Zhu, J. G. (2023). Fabrication of 16 nm Thick Granular L1 0 FePt-hBN Thin Film Media. *IEEE Transactions on Magnetics*, 59(11), 1-5. doi: 10.1109/TMAG.2023.3294354.